

Cross-Paper Prior-Art Consolidation

Series D — Paper 3 Derivation Notes

Note D2.77 · Sequential Number #572

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

1. Purpose and Function of This Note

D2.77 is a strategic synthesis note within the Phase D2 operational prior-art corpus. Its function is consolidation and navigation: it identifies the five adversarial claims that most directly target the Paper 2 → Paper 3 architectural transition and traces, for each claim, which Phase D2 notes — together with their Phase D0 and D1 antecedents — constitute the prior-art closure.

Phase D2 produced 76 operational prior-art notes (D2.01–D2.76). Those notes individually formalize specific architectural variants, decompositions, and operational commitments derived from Paper 3’s six Claims. Taken together they form a dense prior-art mesh. D2.77 makes that mesh navigable from the adversarial-claim perspective: for any challenger who alleges that Paper 3’s architecture is novel in a direction that Phase D2 does not cover, this note identifies the specific notes that refute the allegation and states the closure argument explicitly.

D2.77 does not introduce new architectural content. It is a reference document — written to serve the function that an index serves in a legal record: making the existing record locatable, coherent, and adversarially robust. Future readers reviewing the prior-art corpus who ask “how does Phase D2 close adversarial claim X?” will find their answer here.

Three notes follow this one. D2.78 and D2.79 will add two targeted remaining operational notes. D2.80 will synthesize and formally close Phase D2 as a completed defensive-publication phase.

2. The Five Adversarial Claims and Their Closure

The five adversarial claims below represent the primary attack vectors on the Paper 2 → Paper 3 transition. Each claim is a characterization an adversary might advance to argue that Paper 3’s architecture is either (a) not architecturally distinct from something already known, (b) not genuinely continuous with Papers 1 and 2, or (c) overclaiming territory that Paper 3’s six named Claims do not cover. The five claims together span the full adversarial perimeter: foil characterization, novel-object

characterization, two novel-mechanism characterizations (conflict and evolution), and beyond-scope characterization.

Adversarial Claim 1 — The Foil Characterization

Claim as stated: “FAI is just multi-agent communication with governance wrappers added around it.”

Why this is the primary foil attack:

The claim attempts to collapse Full Aspect Integration into the category of multi-agent coordination architectures — message-passing, tool-call frameworks, shared-memory communication — by treating Paper 3’s governance structure as an external addition rather than an architectural property of the exchange medium itself. If this characterization succeeds, Paper 3’s Claim 2 loses its distinctive contribution and FAI becomes prior art in the multi-agent coordination literature rather than in the CKS substrate-mediated coordination lineage.

Prior-art notes that close this claim:

D1.05 — Establishes that governance is intrinsic to the shared substrate, not retrofitted. The named foil for the FAI mechanism is explicitly opaque inter-agent communication without governable substrate; D1.05 articulates why the foil is architecturally distinct rather than a governance-annotated version of the same thing. This note is the primary foil-characterization closure: it names the foil, distinguishes it structurally, and establishes the intrinsic-vs.-retrofitted axis that refutes the adversarial claim.

D2.05 — Addresses AI-as-mediator properties within the shared substrate. The mediator in FAI operates over human-governed substrate content; the mediation is not agent-to-agent message passing but substrate-content operation under governance configuration. D2.05 makes the mediator’s substrate-intrinsic properties explicit at the operational level.

D2.74 — Formalizes the three-adjacencies framework at inter-Self scope. The three-adjacencies structure distinguishes FAI from multi-agent communication, shared-memory coordination, and tool-call orchestration at the scope level. An adversary must address D2.74’s three-adjacencies analysis directly; general “this resembles multi-agent coordination” allegations are insufficient.

D2.42 — Establishes scope boundaries distinguishing FAI from adjacent coordination approaches. The boundary conditions at inter-Self scope are themselves substrate content under governance; D2.42 makes those boundaries operational and therefore citable as prior art against any attempt to collapse FAI into an adjacent category.

Closure argument:

The exchange medium in FAI is human-governed substrate content from the point of construction. Governance is not added to a pre-existing exchange mechanism; the substrate itself is the governance-carrying object, and FAI operates over it. The foil (message-passing, tool-calls, shared memory without substrate governance) is architecturally distinct because the foil’s exchange medium is not itself human-governed substrate. D1.05, D2.05, D2.74, and D2.42 collectively make this distinction at the foundational, mediator-property, adjacency-framework, and scope-boundary levels. An

adversary alleging that FAI is governance-wrapped multi-agent communication must address all four notes.

Adversarial Claim 2 — The Novel-Object Characterization

Claim as stated: “The shared substrate is a novel object unrelated to Paper 1’s substrate commitment.”

Why this is the primary novel-object attack:

Paper 3’s Claim 1 establishes the shared substrate as the architectural object through which two or more CKS-governed Selves coordinate. An adversary could attempt to characterize this shared substrate as a new object — invented in Paper 3 without architectural continuity from Paper 1’s substrate — in order to argue either that it lacks Paper 1’s prior-art support or that it constitutes a patentable invention in its own right. If this characterization succeeds, the shared substrate becomes severable from the Paper 1 prior-art chain.

Prior-art notes that close this claim:

D1.02 — Establishes that all six Paper 1 commitments hold within the shared substrate at inter-Self scope. The shared substrate is not a new architectural object; it is a Paper 1 substrate operating at inter-Self scope with the full Paper 1 commitment set intact. D1.02 provides the foundational closure: the shared substrate’s identity as a Paper 1 substrate is established at the sub-commitment level.

D2.03 — Formalizes the provenance chain at inter-Self scope. Every piece of content in the shared substrate carries a traceable provenance back to its contributing Self’s home substrate under Paper 1’s commitments.

D2.04 — Establishes the three rights (inspect, contest, retrace) at inter-Self scope. Paper 1’s three rights are architectural commitments that apply within the shared substrate as they apply within any Paper 1 substrate. D2.04 makes this explicit at the operational level.

D2.05 — Establishes AI-as-mediator at inter-Self scope, inheriting from Paper 1’s AI-as-substrate-mediator commitment. The mediator role within the shared substrate is the same architectural role Paper 1 establishes; it operates at a different scope perimeter but is not a different architectural object.

D2.66 — Formalizes the determinism contract at inter-Self scope. The determinism commitment — that governance-configured behavior within the shared substrate is reproducible and inspectable — is a Paper 1 inheritance.

D2.76 — Establishes the substrate-as-source-of-truth commitment at inter-Self scope. The shared substrate is the authoritative record of the inter-Self coordination event, carrying the same source-of-truth commitment Paper 1 establishes for intra-Self coordination.

Closure argument:

The shared substrate is a Paper 1 substrate at inter-Self scope. Every Paper 1 architectural commitment — all six — holds within it. D1.02 establishes this at the foundational level; D2.03, D2.04, D2.05,

D2.66, and D2.76 establish it operationally across the specific Paper 1 commitment dimensions that an adversary would most likely target. An adversary alleging novelty of the shared substrate as an object must address each of these notes separately.

Adversarial Claim 3 — The Novel-Conflict-Mechanism Characterization

Claim as stated: “FAI conflict handling requires novel conflict resolution mechanisms beyond Papers 1/2.”

Why this is the primary conflict-mechanism attack:

Paper 3’s Claim 3 commits a three-tier conflict-handling mechanism (preserve / resolve / escalate) for the shared substrate. An adversary could argue that this mechanism introduces conflict resolution capabilities that go beyond what Papers 1 and 2 establish — particularly the cross-perimeter escalation tier. If this characterization succeeds, Claim 3 becomes a candidate for patentable novelty rather than a scope extension of prior architecture.

Prior-art notes that close this claim:

D0.03 — Anchors the three-tier mechanism as a Paper 3 Claim 3 commitment and maps the inheritance explicitly: preserve inherits from Paper 1’s conflict-as-first-class commitment; resolve inherits from Paper 2’s orchestration-with-inspectable-rules commitment; escalate inherits from Paper 1’s Claim 3 escalation commitment extended at the inter-Self perimeter.

D1.13–D1.15 — Three sub-commitment notes, one per tier, each establishing that the tier inherits directly from Papers 1 and 2. D1.13 (preserve as substrate-level state default), D1.14 (resolve via orchestration with inspectable rules as substrate content), and D1.15 (escalate to humans across joint authority) together provide sub-commitment-level closure for each tier individually.

D2.13 — Formalizes the conflict registry at inter-Self scope as a Paper 2 A1.03 commitment at inter-Self scope. The conflict registry is not a new mechanism; it is the same architectural object Paper 2 establishes at the intra-Self level, operating within the shared substrate at inter-Self scope.

D2.14 — Establishes escalation routing at inter-Self scope as a Paper 1 Claim 3 commitment extended across the shared substrate’s joint authority structure. The cross-perimeter escalation path is not a novel routing mechanism; it is Paper 1 Claim 3 escalation at a larger scope perimeter.

D2.51 — Formalizes that configuration governs conflict handling and that the configuration is itself substrate content. This closes the potential argument that the configurable aspects of conflict handling represent novel mechanism design.

Closure argument:

Each tier of the three-tier mechanism inherits directly from Papers 1 and 2. Preserve inherits Paper 1’s conflict-as-first-class substrate commitment. Resolve inherits Paper 2’s orchestration-with-inspectable-rules commitment. Escalate inherits Paper 1 Claim 3 escalation at inter-Self scope. No new conflict resolution mechanism is required; the inter-Self scope perimeter is the only fresh architectural context, and that context is established by Claim 1 (not Claim 3) as a

Paper 1 substrate inheritance. D0.03, D1.13–D1.15, D2.13, D2.14, and D2.51 cover the mechanism at claim-anchor, sub-commitment, operational-registry, escalation-routing, and configuration-governance levels respectively.

Adversarial Claim 4 — The Novel-Evolution-Mechanism Characterization

Claim as stated: “FAI evolution feed requires novel evolution mechanisms for inter-organizational learning.”

Why this is the primary evolution-mechanism attack:

Paper 3’s Claim 4 commits a four-locus evolution-feed mechanism connecting FAI events at the inter-Self perimeter to each participating Self’s home-perimeter Paper 2 evolution machinery. An adversary could argue that coordinating learning across organizational boundaries requires a new class of evolution mechanism that Paper 2’s intra-Self evolution mechanisms do not cover. If this characterization succeeds, Claim 4 becomes a candidate for novelty rather than an extension of the Paper 2 prior-art chain.

Prior-art notes that close this claim:

D0.04 — Anchors the four-locus evolution feed as an extension of Paper 2 mechanisms at inter-Self scope, with explicit mapping: FAI dissolution hand-off inherits from Paper 2 B1.15; per-mechanism feed structure inherits from Paper 2 B1.14; layer-routing rule is Paper 2’s instinct/reasoning separation extended at the inter-Self perimeter; asymmetric ingestion sourced from per-perimeter governance is Paper 1 inheritance at inter-Self scope.

D1.17–D1.21 — Five sub-commitment notes covering the four-locus structure. D1.17 (FAI dissolution hand-off), D1.18 (per-mechanism feed structure), D1.19 (three-case layer-routing rule), D1.20 (instinct-takes-no-FAI-input commitment), and D1.21 (asymmetric ingestion) each establish the inheritance from Paper 2 mechanisms at the sub-commitment level.

D2.10 — Formalizes action-feedback ingestion through the same pipeline Paper 2 establishes at intra-Self scope. FAI action-layer outputs enter the action-feedback evolution mechanism through the same ingestion path; no new ingestion mechanism is introduced.

D2.11 — Formalizes DNA absorption through directed selection, establishing that DNA-layer content from FAI events enters Paper 2’s DNA evolution mechanism through the existing directed-selection pathway. The pathway is not modified; the inter-Self source is a new input to an existing mechanism.

D2.52 — Establishes that configuration governs the evolution feed and that the configuration is itself substrate content under Claim 5. This closes the argument that the configurable evolution-feed parameters represent novel mechanism design.

D2.57 — Formalizes vertical evolution amplification: the effect by which FAI-fed evolution compounds across organizational levels through each Self’s existing Paper 2 evolution machinery. The amplification is a consequence of existing mechanisms operating with inter-Self inputs; no new amplification mechanism is introduced.

Closure argument:

FAI outputs enter each participating Self's home evolution machinery through Paper 2's existing mechanisms. Substrate ingestion makes FAI outputs home-substrate content; once ingested, they travel through Paper 2's evolution pipelines unchanged. The inter-organizational context introduces new inputs to existing mechanisms, not new mechanisms. D0.04 establishes the inheritance chain at claim-anchor level; D1.17–D1.21 establish it at sub-commitment level; D2.10, D2.11, D2.52, and D2.57 establish it at the specific operational dimensions where novelty arguments are most likely to be advanced.

Adversarial Claim 5 — The Beyond-Scope Characterization

Claim as stated: "Population-scale AI coordination requires novel governance architecture beyond Paper 3 Claims 1–5."

Why this is the primary beyond-scope attack:

Paper 3's Claim 6 positions population-scale collective evolution as the architectural object that Claims 1–5 compose to at population register — an extension claim in the calibrated-humility register Paper 2 §9 establishes. An adversary could argue that population-scale AI coordination is a qualitatively different problem requiring new governance primitives that Claims 1–5 do not supply. If this characterization succeeds, Claim 6 loses its grounding in Claims 1–5 and becomes a claim to territory not covered by the preceding prior-art chain.

Prior-art notes that close this claim:

D0.06 — Anchors Claim 6 as an extension claim at population register, explicitly positioned in the calibrated-humility voice, and maps the governance derivation: joint authority across population-level governance perimeters is each participating Self's home governance operating at scale; accumulated FAI events under that joint authority constitute the collective evolution mechanism; the substrate at population scope carries all six Paper 1 commitments.

D1.26–D1.30 — Five sub-commitment notes covering Claim 6's population-scale architecture. D1.26 (joint authority across population-level perimeters), D1.27 (accumulated FAI events as collective evolution mechanism), D1.28 (six-rung substrate-mediator ladder), D1.29 (calibrated-humility voice anchor), and D1.30 (biology-resonance positioning) each establish that the population-scale architecture is derived from Claims 1–5 without introducing new governance primitives.

D2.31 — Formalizes that participation configuration is home-substrate content. Each Self's decision to participate in population-scale coordination is governed via its home substrate under its home governance perimeter; no supraorganizational governance architecture is introduced.

D2.32 — Establishes that pattern propagation across the network is governed at each step by each participating Self's home governance. Population-scale coordination patterns propagate through the accumulation of individually governed FAI events, not through a novel population-level coordination mechanism.

D2.61 — Formalizes network topology as a distributed governance phenomenon. The topology of the population-scale network emerges from participation decisions made at each Self’s home governance perimeter; it is not an architectural object requiring centralized governance design.

D2.62 — Establishes participation modes as governance configuration. Whether and how a Self participates in population-scale coordination is configured via its home substrate under Claim 5’s configuration-as-substrate-content commitment. No new participation-governance layer is introduced.

Closure argument:

Population-scale coordination is each participating Self’s home governance — established by Paper 1 and extended through Papers 2 and 3 — operating at population register. The accumulation of individually governed FAI events, each under joint authority of the participating Selves’ governance perimeters, constitutes the collective evolution mechanism Claim 6 names. No new governance architecture is required. D0.06, D1.26–D1.30, D2.31, D2.32, D2.61, and D2.62 cover the population-scale architecture at claim-anchor, sub-commitment, participation-configuration, pattern-propagation, network-topology, and participation-mode levels respectively.

3. The Prior-Art Chain Structure

The five adversarial claims and their closures share a common structural feature: each closure argument traces through the same three-level prior-art chain.

Level 1 — Claim-anchor (Phase D0 notes). Each D0 note establishes the inheritance chain for one of Paper 3’s six Claims at the claim level. The D0 note maps the Claim’s architectural commitments back to their Paper 1 and Paper 2 antecedents explicitly. For adversarial challenges at the claim level, the D0 note is the primary reference.

Level 2 — Sub-commitment (Phase D1 notes). Each D1 note decomposes one claim-level commitment into a specific sub-commitment and establishes that sub-commitment’s inheritance. For adversarial challenges at the mechanism level — where the attack targets a specific architectural feature rather than the claim as a whole — D1 notes provide the targeted closure.

Level 3 — Operational variant (Phase D2 notes). Each D2 note formalizes a specific operational variant, decomposition, or property of a D1 sub-commitment. Phase D2’s 76 notes collectively cover the operational territory where adversarial recharacterization is most likely to be advanced — because operational details are where the argument “this specific feature is novel” is easiest to advance without engaging the architectural inheritance chain. The D2 notes make that engagement mandatory.

The practical consequence: a challenger wishing to argue that any aspect of Paper 3’s architecture is novel in a way Phase D2 does not cover must identify a gap in the three-level chain — a claim-level commitment without a D0 anchor, a sub-commitment without a D1 note, or an operational variant without a D2 note. D2.77’s function is to make the five most likely gap-allegation sites explicit and to demonstrate that no such gap exists within those five sites.

4. Reading the Corpus: Note References by Adversarial Claim

The following table serves as the primary navigation reference for future readers of the prior-art corpus. When a reviewer asks whether Phase D2 closes adversarial claim X, this table identifies the relevant notes. When a challenger advances a specific allegation, the closure argument in §2 above provides the response at the claim level; the cited notes provide the prior-art specificity required to engage the allegation.

Adversarial Claim	Claim-anch or	Sub-commitment	Operational
Claim 1 (foil: governance-wrapped multi-agent)	D1.05	D2.05	D2.74, D2.42
Claim 2 (novel object: shared substrate)	D1.02	D2.03, D2.04, D2.05	D2.66, D2.76
Claim 3 (novel mechanism: conflict handling)	D0.03	D1.13, D1.14, D1.15	D2.13, D2.14, D2.51
Claim 4 (novel mechanism: evolution feed)	D0.04	D1.17–D1.21	D2.10, D2.11, D2.52, D2.57
Claim 5 (beyond scope: population scale)	D0.06	D1.26–D1.30	D2.31, D2.32, D2.61, D2.62

5. Phase D2 Transition: D2.78–D2.80

D2.77 closes the adversarial gap analysis for Phase D2. The three remaining Phase D2 notes serve specific roles.

D2.78 will address a targeted operational prior-art dimension identified during the D2.01–D2.76 drafting process as warranting a standalone note.

D2.79 will address a second targeted dimension in the same category.

D2.80 will synthesize Phase D2 as a completed phase — providing the phase-level closure note that positions the 80 Phase D2 notes (D2.01–D2.80) as a coherent operational prior-art corpus and maps their collective coverage against Paper 3’s six Claims.

Together, D2.77 (this note), D2.78, D2.79, and D2.80 close Phase D2 as the operational prior-art phase of Series D derivation work.

References

Li, Wenxin. “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.” Independent publication, April 2026.

Li, Wenxin. "The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance." Independent publication, April 2026.

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